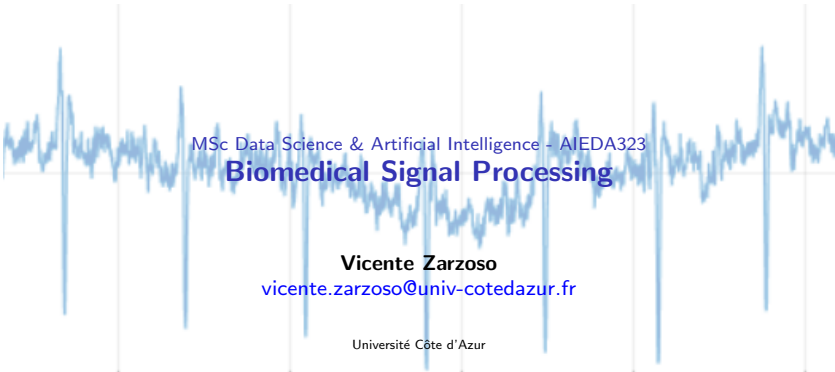




MSc Data Science & Artificial Intelligence - AIEDA323

Biomedical Signal Processing

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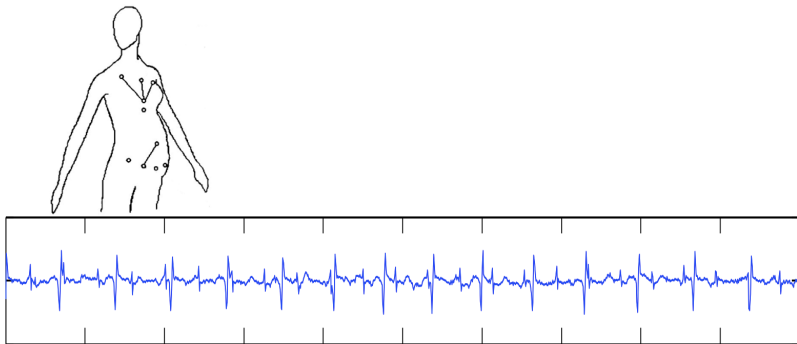
Chapter 2: Adaptive noise canceling

Objectives

- Realize the **limitations of frequency filtering** for signal denoising
- Understand the interest of **spatial filtering** for exploiting **spatial diversity**
 - ▶ biomedical signal acquisition systems are often composed of several spatially separated sensors (e.g., 12-lead ECG)
- Study a **classical spatial filtering** technique
 - ▶ adaptive noise canceling (ANC)
- Understand its **model assumptions and limitations**
- Derive **algebraic** (closed-form) **and iterative solutions**
 - ▶ optimal learning step selection
- **Implement** ANC using **MATLAB**
- Illustrate its performance in **biomedical signal processing problems** [lab assignment]

Motivating example 1: Fetal ECG extraction

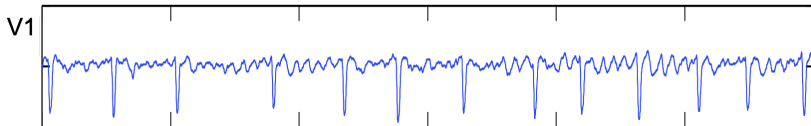
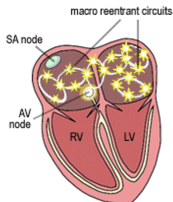
- Fetal ECG (FECG) is corrupted by maternal ECG (MECG) in surface records during pregnancy



- How to process the available signals optimally to cancel the MECG and estimate the FECG?

Motivating example 2: Atrial activity extraction

- Atrial activity (AA) is corrupted by ventricular activity (VA) in atrial fibrillation (AF) ECGs



- How to process the available signals optimally to cancel the VA and estimate the AA?

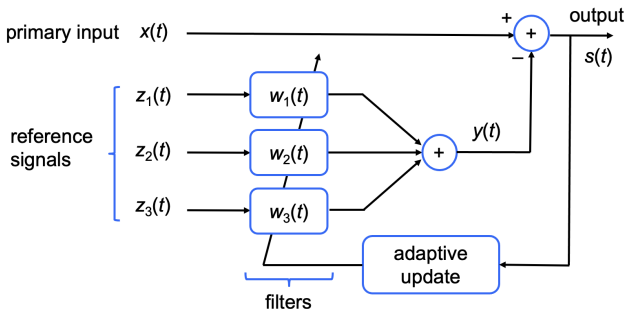
→ **Exercise 1:** A frequency filtering approach may generally not be suitable in the above examples. Why?

Adaptive noise canceling (ANC)

- Several simultaneous records are available (e.g., multiple ECG leads)
 - ▶ FECG extraction: abdominal and thoracic maternal leads
 - ▶ AA extraction: standard ECG (12 leads), BSPM
- Some leads don't contain contributions from the signal or interest (SOI) but only from the interference
SOI
 - ▶ FECG extraction:
 - ★ abdominal leads: FECG + MECG
 - ★ thoracic leads: MECG only
 - ▶ AA extraction:
 - ★ some leads (e.g., V1): significant AA contribution + VA
 - ★ other leads: mainly VA
- First proposed by Widrow and co-workers in the 1970s.¹

¹B. Widrow, J. R. Glover, J. M. McCool, "Adaptive noise cancelling: principles and applications", *Proceedings of the IEEE*, 63(12):1692-1716, 1975.

ANC — modeling assumptions



- Primary input: $x(t) = d(t) + \underline{b(t)}$

- ▶ $d(t)$: signal of interest (SOI), desired signal
- ▶ $b(t)$: interference, decorrelated with the SOI

- Reference signals: $\{z_r(t)\}_{r=1}^R$

- ▶ decorrelated with the SOI: $E\{z_r(t)d(t)\} = 0$
- ▶ correlated with the interference: $E\{z_r(t)b(t)\} \neq 0$

- Estimated interference:

$$y(t) = \sum_{r=1}^R w_r(t) * z_r(t)$$

- Output signal:

$$s(t) = x(t) - y(t)$$

Optimization criterion

- **Goal 1:** find filter impulse responses $\{w_r(t)\}_{r=1}^R$ making $s(t)$ as close as possible to $d(t)$.

→ *Exercise 2:* Prove that goal 1 is equivalent to

- **Goal 2:** find filter impulse responses $\{w_r(t)\}_{r=1}^R$ making $y(t)$ as close as possible to $b(t)$.

→ *Exercise 3:* Prove that goal 2 is equivalent to

- **Goal 3:** find filter impulse responses $\{w_r(t)\}_{r=1}^R$ minimizing $E\{s^2\}$.

→ *Exercise 4:* What is the advantage of goal 3 over goals 1 and 2?

Optimization criterion (cont'd)

Problem statement

Given the observed primary input $x(t)$ and reference signals $\{z_r(t)\}_{r=1}^R$, find filter impulse responses $\{w_r(t)\}_{r=1}^R$ minimizing

$$\xi = E\{s(t)^2\}$$

Particular case: M th-order FIR filters

$$w_r(t) * z_r(t) = \sum_{k=0}^M w_r(k) z_r(t-k) = \mathbf{w}_r^T \mathbf{z}_r(t)$$

$$\mathbf{w}_r \stackrel{\text{def}}{=} [w_r(0), w_r(1), \dots, w_r(M)]^T \in \mathbb{R}^{M+1} \quad \mathbf{z}_r(t) \stackrel{\text{def}}{=} [z_r(t), z_r(t-1), \dots, z_r(t-M)]^T \in \mathbb{R}^{M+1}$$

$$y(t) = \sum_{r=1}^R w_r(t) * z_r(t) = \sum_{r=1}^R \mathbf{w}_r^T \mathbf{z}_r(t) = \mathbf{w}^T \mathbf{z}(t)$$

$$\mathbf{w} \stackrel{\text{def}}{=} [\mathbf{w}_1^T, \mathbf{w}_2^T, \dots, \mathbf{w}_R^T]^T \in \mathbb{R}^{R(M+1)} \quad \mathbf{z}(t) \stackrel{\text{def}}{=} [\mathbf{z}_1(t)^T, \mathbf{z}_2(t)^T, \dots, \mathbf{z}_R(t)^T]^T \in \mathbb{R}^{R(M+1)}$$

Optimization criterion

$$\mathbf{w}_* = \arg \min_{\mathbf{w}} \xi \quad \xi = E\{s(t)^2\} \quad s(t) = x(t) - \mathbf{w}^T \mathbf{z}(t)$$

Optimal algebraic solution: Wiener-Hopf

$$\nabla \xi = -2E\{s(t)z(t)\} = 0$$

$$E\{sz\} = E\{(x - z^T w)z\} = \underbrace{E\{xz\}}_{r_{xz}} - \underbrace{E\{zz^T\}}_{R_{zz}} w = r_{xz} - R_{zz} w$$

$$\nabla \xi = 2(R_{zz} w - r_{xz}) = 0$$

Wiener-Hopf solution

$$w_* = R_{zz}^{-1} r_{xz}$$

Drawbacks

- Requires inversion of $R_{zz} \in \mathbb{R}^{R(M+1) \times R(M+1)}$
 - ▶ high computational cost
 - ▶ numerical stability problems
- Unable to track input signal variations.

Iterative solution: batch gradient descent

Principle: move in little steps in the direction opposite the gradient

Gradient descent

$$\left\{ \begin{array}{l} \mathbf{w}^{(i+1)} = \mathbf{w}^{(i)} - \mu \nabla \xi(\mathbf{w}^{(i)}) \\ \mathbf{w}^{(i+1)} = \mathbf{w}^{(i)} - 2\mu(\mathbf{R}_{zz}\mathbf{w}^{(i)} - \mathbf{r}_{xz}) \end{array} \right. \quad \checkmark$$

$\mathbf{w}^{(i)}$: filter coefficients at i th iteration

μ : *adaption step or learning coefficient*

In practice, finite signal length $\rightarrow$ replace ensemble statistics (expectations) by sample estimates

Batch gradient: expectation is replaced by *sample mean* over available time samples

$$\checkmark \quad \hat{\mathbf{R}}_{zz} = \frac{1}{N} \sum_{t=0}^{N-1} \mathbf{z}(t)\mathbf{z}(t)^T \quad \hat{\mathbf{r}}_{xz} = \frac{1}{N} \sum_{t=0}^{N-1} x(t)\mathbf{z}(t)$$

Drawback: Unable to track input signal variations within processed sample window.

Iterative solution: stochastic gradient descent

Principle

- Drop expectation operator $\rightarrow$ one-sample estimate of the gradient
- Update filter vector with each new input sample

$$\nabla \xi(\mathbf{w}) = 2(\mathbf{R}_{zz}\mathbf{w} - \mathbf{r}_{xz}) = 2(\mathbf{E}\{\mathbf{z}\mathbf{z}^T\}\mathbf{w} - \mathbf{E}\{x\mathbf{z}\}) \approx 2(\mathbf{z}(t)\mathbf{z}(t)^T\mathbf{w} - x(t)\mathbf{z}(t)) = 2\mathbf{z}(t)(\mathbf{z}(t)^T\mathbf{w} - x(t))$$

Stochastic gradient descent (SGD)

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - 2\mu\mathbf{z}(t)(\mathbf{z}(t)^T\mathbf{w}^{(t)} - x(t))$$

This update rule is known as **adaptive noise canceling (ANC)**.

With multiple ($R > 1$) reference signals: **multi-reference ANC (MRANC)**.

Q: How to choose the step size μ to guarantee convergence?

Learning step selection

Consider the gradient descent algorithm based on ensemble statistics:

$$\mathbf{w}^{(i+1)} = \mathbf{w}^{(i)} - 2\mu(\mathbf{R}_{zz}\mathbf{w}^{(i)} - \mathbf{r}_{xz})$$

Define

$$\Delta\mathbf{w}^{(i)} = \mathbf{w}^{(i)} - \mathbf{w}_* \quad (1)$$

Then:

$$\Delta\mathbf{w}^{(i+1)} = \Delta\mathbf{w}^{(i)} - 2\mu(\mathbf{R}_{zz}(\Delta\mathbf{w}^{(i)} + \mathbf{w}_*) - \mathbf{r}_{xz}) = \Delta\mathbf{w}^{(i)} - 2\mu(\mathbf{R}_{zz}\Delta\mathbf{w}^{(i)} + \underbrace{\mathbf{R}_{zz}\mathbf{w}_* - \mathbf{r}_{xz}}_0)$$

$$\Delta\mathbf{w}^{(i+1)} = (\mathbf{I} - 2\mu\mathbf{R}_{zz})\Delta\mathbf{w}^{(i)} = (\mathbf{I} - 2\mu \underbrace{\mathbf{U}\mathbf{\Lambda}\mathbf{U}^T}_{\text{EVD of } \mathbf{R}_{zz}})\Delta\mathbf{w}^{(i)} \stackrel{\substack{\uparrow \\ \mathbf{U}\mathbf{U}^T = \mathbf{I}}}{=} \mathbf{U}(\mathbf{I} - 2\mu\mathbf{\Lambda})\mathbf{U}^T\Delta\mathbf{w}^{(i)}$$

Multiply both sides of the above expression by $\mathbf{U}^T$ and define

$$\Delta\tilde{\mathbf{w}}^{(i)} = \mathbf{U}^T\Delta\mathbf{w}^{(i)} \quad (2)$$

Then:

$$\Delta\tilde{\mathbf{w}}^{(i+1)} = (\mathbf{I} - 2\mu\mathbf{\Lambda})\Delta\tilde{\mathbf{w}}^{(i)} = (\mathbf{I} - 2\mu\mathbf{\Lambda})^{i+1}\Delta\tilde{\mathbf{w}}^{(0)}$$

Learning step selection (cont'd)

$$\Delta \tilde{\mathbf{w}}^{(i+1)} = (\mathbf{I} - 2\mu\mathbf{\Lambda})^{i+1} \Delta \tilde{\mathbf{w}}^{(0)}$$

Matrix $(\mathbf{I} - 2\mu\mathbf{\Lambda})$ is diagonal, with diagonal entries $(1 - 2\mu\lambda_k)$, where $\{\lambda_k\}_{k=1}^{R(M+1)}$ denote the eigenvalues of $\mathbf{R}_{zz}$.

From eqns. (1)–(2), $\Delta \tilde{\mathbf{w}}^{(i)} = \mathbf{U}^T(\mathbf{w}^{(i)} - \mathbf{w}_*)$, and therefore:

$$\mathbf{w}^{(i)} \xrightarrow{i \rightarrow \infty} \mathbf{w}_* \iff \Delta \tilde{\mathbf{w}}^{(i)} \xrightarrow{i \rightarrow \infty} \mathbf{0}$$

Assuming that $\Delta \tilde{\mathbf{w}}^{(0)} \neq \mathbf{0}$, convergence occurs if and only if

$$|1 - 2\mu\lambda_k| < 1, \quad \forall k$$

leading to

ANC convergence condition

$$0 < \mu < \frac{1}{\lambda_{\max}}$$

Learning step selection (cont'd)

$$\lambda_{\max} < \sum_{k=1}^{R(M+1)} \lambda_k = \text{trace}(\mathbf{R}_{zz}) = (M+1) \sum_{r=1}^R \mathbb{E}\{z_r^2\}$$

Alternative convergence condition

$$0 < \mu < \frac{1}{(M+1) \sum_{r=1}^R \mathbb{E}\{z_r^2\}}$$

- It can be directly determined from the reference signals without explicit computation of the correlation matrix $\mathbf{R}_{zz}$ and its EVD.
- More conservative criterion than that based on $\lambda_{\max}$.

Algorithm implementation

→ Exercise 5:

① Create a function `mranc.m` implementing the SGD-MRANC algorithm

► **Input parameters**

primary input	row vector $\mathbf{x}$ with length N
reference signals	matrix $\mathbf{Z}$ with dimensions $R \times N$ (one signal per row)
filter order	scalar M
adaption coefficient	scalar μ

► **Output parameters**

output signal	row vector $\mathbf{s}$ with length N
estimated filters	matrix $\hat{\mathbf{W}}$ with dimensions $R(M+1) \times N$ (one vector $\mathbf{w}^{(i)}$ per column)

② Generate a synthetic signal set:

- Desired signal $d(t)$: unit-amplitude sinusoid with frequency 5 Hz and random initial phase, sampled at a rate of 500 Hz during 2 seconds.
- Primary signal $x(t)$: desired signal contaminated by a sinusoid of 50 Hz and random phase (mains interference), signal-to-interference ratio SIR = -10 dB.
- Reference signals $z_1(t)$ and $z_2(t)$: unit-amplitude sinus and cosinus with the same frequency as the mains interference and zero initial phase.

③ With the aid of function `mranc.m`, obtain the MRANC filtered signal with $M = 0$ and $\mu = 0.01$. Plot the desired signal, the primary input and the output signal, as well as the error (difference between the desired and the output signal).

Algorithm implementation (cont'd)

→ *Exercise 6*: Evaluate the desired signal estimation quality in the results of the previous exercise:

- ① Plot the obtained filter coefficients and compare them with their expected theoretical values.
- ② Compute, plot and compare the frequency spectrum (dB) of the desired signal, the primary input and the output signal. For the latter, consider only the segment of samples after convergence of the adaptive algorithm.
- ③ Compute and compare the power of the desired signal, the primary input, the output signal and the error signal. Consider only the signal portion estimated after convergence.
- ④ Estimate the output SIR and the SIR improvement brought about by the MRANC algorithm.

Implementation tip: avoid innermost loop by using a Toeplitz matrix

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - 2\mu \mathbf{z}(t) (\mathbf{z}(t)^T \mathbf{w}^{(t)} - x(t))$$

If $R = 1$:

$$\mathbf{w}^{(t)} = [w(0)^{(t)}, w(1)^{(t)}, \dots, w(M)^{(t)}]^T \in \mathbb{R}^{M+1} \quad \mathbf{z}(t) = [z(t), z(t-1), \dots, z(t-M)]^T \in \mathbb{R}^{M+1}$$

$$\mathbf{z}(t)^T \mathbf{w}^{(t)} = \mathbf{w}^{(t)T} \mathbf{z}(t)$$

$$\mathbf{Z}_M \stackrel{\text{def}}{=} [\mathbf{z}(0), \mathbf{z}(1), \dots, \mathbf{z}(N-1)] = \begin{bmatrix} z(0) & z(1) & z(2) & z(3) & \cdots & z(N-1) \\ 0 & z(0) & z(1) & z(2) & \ddots & z(N-2) \\ 0 & 0 & z(0) & z(1) & \ddots & z(N-3) \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \cdots & z(N-M-1) \end{bmatrix} \in \mathbb{R}^{(M+1) \times N}$$

$$\mathbf{w}^{(t)T} \mathbf{z}(t) = \mathbf{w}^{(t)T} [\mathbf{Z}_M]_{:,t+1}$$

- Toeplitz matrix $\mathbf{Z}_M$ can be generated from the reference signals (input parameter $\mathbf{Z}$) before starting the iterations over signal samples.
- Loop over entries of $\mathbf{w}^{(t)}$ and $\mathbf{z}(t)$ is replaced by the scalar product of $\mathbf{w}^{(t)}$ and the $(t+1)$ th column of $\mathbf{Z}_M \rightarrow$ fast MATLAB implementation.
- If $R > 1$, matrix $\mathbf{Z}_M \in \mathbb{R}^{R(M+1) \times N}$ is block Toeplitz and can be constructed by stacking the Toeplitz matrix of each reference signal.

Performance evaluation

• Synthetic signals

- ▶ always start by generating “toy problem” signals → perfectly known ground truth
- ▶ run the algorithm in controlled conditions to make sure it works properly before applying it to real data
- ▶ understand the effect of different parameters on algorithm performance
 - ★ adaption coefficient μ
 - ★ filter order M
- ▶ test the effect of deviations from model assumptions
 - ★ desired signal leakage into reference leads
- ▶ generate random realizations to evaluate performance distribution with respect to modified parameters (Monte Carlo simulations)

• Real recordings

- ▶ once the algorithm has been validated on synthetic signals, it can be applied to real data
- ▶ carefully choose algorithm parameters for optimal results

Summary

- Frequency filtering is not useful if spectra of desired signal and interference overlap
- Multiple spatially separated sensors can be exploited by spatial filtering techniques
- ANC
 - ▶ desired signal is obtained by filtering reference signals and subtracting the estimated interference from the primary input
 - ▶ minimizing output signal power = minimizing error between output signal and desired signal
 - ▶ reference signals must be correlated with the interference, uncorrelated with the desired signal
- Wiener-Hopf solution is optimal but often impractical
 - ▶ large matrix inversion, computational cost, numerical instability
- Iterative gradient-descent algorithms
 - ▶ low complexity, learning step selection guaranteeing convergence
- Main limitation:
 - ▶ model assumptions impose constraints on sensor placement